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APPLICATION OF TIME STUDY TECHNIQUE- A CASE STUDY OF ALLOY WHEEL INDUSTRY

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***Abstract*- Over a time economic competitiveness has changed globally. The many countries have joined the global economic competition to take global market in order to make profit and tough competition to their competitors by increasing productivity, for this purpose they have adapted various techniques. There are many product and process variations in production system; to increasing their productivity various methods or techniques are available in market. The motive for this study was increment in output over a given time span putting in another ways, improving productivity. There are different industrial engineering techniques amongst which time study techniques are using for productivity improvement tools. This research searched for bottlenecks using time study which was inspected at degating, rough boring and hole drilling work stations. Possible solutions and alternative aimed at achieving the objectives. This study is best suited for this alloy wheel manufacturing industry as it reduces the unwanted time.**

Keywords- Time Study, Industrial Engineering Tools, Time Study Sheet, Standard time observation table

I. INTRODUCTION

In this paper study of ABC alloy wheel manufacturing company which is generally a small scale industry. In this company manufacturing an alloy wheel with the help of casting and also involves many machining process to finishing the wheel. So here time study technique applied on the all machining processes to find out the standard time of machining operations.

Time Study is an important technique mostly used in industries for improving the productivity of plant. Time study provides standard time for each and every ongoing operation. It gives suitable time for each operation which is used to increase productivity. Productivity is basically a function of revenue generated by a company during its financial year. Time study is what actually helps increasing the company outcomes by reducing manufacturing time, which indirectly increases productivity.

Basic instruments used for time study are stop watch, study board, Data sheet etc. Basically in time study each operation is divided into number of elements to identify which is human work part and which is the machine work part. So it becomes easy to add allowances and to calculate standard time.

Generally with the help of time study the bottlenecks in process can be found out and the reduction in the operation time takes place. it means the elimination of non valuable time by improving the process with the suggested changes.(1)

In industry with help of every small improvements and eliminating unwanted cycle time increases productivity of system. It also gives standardization to process and improve man power efficiency which really helps to increase productivity. (2)

II. METHODOLOGY

In this paper, it was selected as a case study the cycle time of an alloy wheel manufacturing processes. Following procedure are to be performed for time study in various operations :(3)

1. Detailed study of each operation
2. Making flow chart on each processes in sequence manner
3. Starting the time study by selecting sequenced operation and note observed time of each operation for 20 cycles
4. Giving rating to the operator as per observation and operator skills for each cycle of operation
5. Then calculating normal time of each cycle of operation.

   Normal time= Observed time x Rating Factor
6. Considering allowances (i.e. personal need, fatigue and contingency allowances)
7. Calculating standard time

   Standard time= Normal time + allowances
8. Plot all standard time in data sheet.

The data in time study is recorded by using time study sheet, this sheet contains data related to the operation like element description, percentage of rating, watch reading, subtracted time and basic time. Time study sheet shown in below fig.1

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Fig.1 Data recorded time study sheet (3)

Flow chart is shown below fig.3 according to departments

Foundry Department

(Casting, Degating, Heat Treatment, Ageing)

⇓

Machining Department

(Rough Boring, 1st A Cut, 1st B Cut, Inner cut, Air Hole, Hole Drilling)

⇓

Quality Department

(Air Leak Test, Wheel Balancing, Inspection)

⇓

Finishing Department

(Buffing, Short Blasting, Washing, Ovening, Face Cutting)

⇓

Packaging Department

(Packaging, Storage)

Fig.2 Flow Chart

III. DATA ANALYSIS:

After the performance of time study in various operations of alloy manufacturing company results are recorded in data sheets. These data contains standard time of each operation. Considering allowances as 15% of basic time.

Operations	Avg. Basic Time(mins)	Standard Time(mins)
Casting	7.8	8.96
Degating	1	1.15
Heat Treatment	240	278
Ageing	61	70.15
Rough Boring	2.33	2.67
1st A Cut	1.36	1.56
1st B Cut	1.25	1.44
Inner Cut	1.3	1.49
Hole Drilling	5.6	6.44
Air Leak Test	1.77	2.04
Wheel Balancing	0.84	0.97
Inspection	0.85	0.97
Air Hole	1.03	1.18
Buffing	1.7	1.96
Short Blasting	0.87	1
Washing	1.5	1.72
Ovening	1.8	2.07
Face Cut	1.3	1.495
Washing	1.5	1.72
Packaging	1	1.15
	Total Time	**388 (mins)**

Table No. 1

On the basis table no.1 data pie chart gives the systematic representation of standard time of various operation.

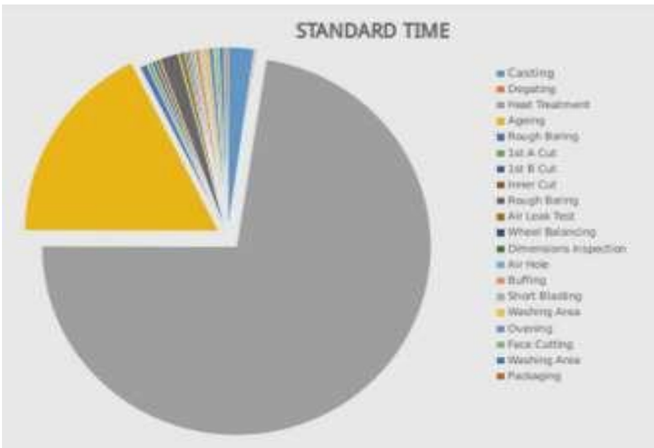

Fig.3 Pie chart of standard time

Heat treatment and Ageing are the most time consuming process as seen in the Data table as well as Fig.3. Reducing time with these processes will lead to compromise in durability and quality of the product. Although its time consuming, these processes are carried out for 100 quantities at a time so saving time here is not affected to the total basic time as much as it would affect by targeting the other processes. Thus targeting the other processes involved in the company to reduce time by analyzing the shortcomings with respect to each processes will lead to most beneficial standard time for production of each alloy wheel in production.

IV. SUGGETIONS

Table no.1 shows that rough boring and hole drilling consumes maximum time in comparison with other machining activities. So, the focus is on activities which are consuming more time in comparison with other elements.

Consequently these elements were given greater emphasis in the discussion section.

a) Using multiple air hole drilling

Currently hole drilling is performed using single spindle head and it consumes time to drill multiple holes. So to reduce this extra time using their a multiple spindle (gang drilling) head. If using a gang drilling technique saving of time is approximately 40% to 50%.

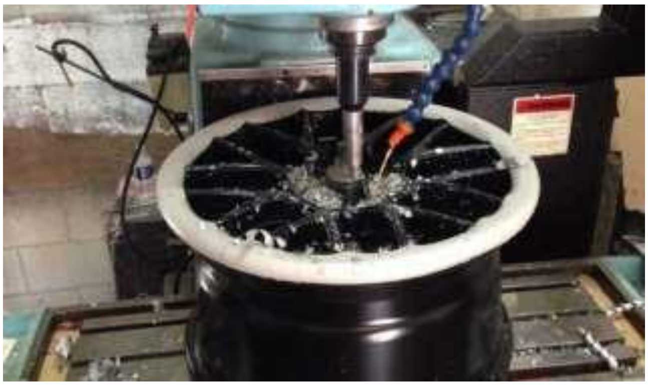

Fig.4 Hole Drilling Operation

b) Combining of two operation

Here table no. 1 showing the degating operation is performed before the rough boring operation and for operation of degating using CNC machine but rough boring operation is performed on semi automatic lathe machine. If they combining the rough boring operation with degating operation, so performing both operations on CNC machine it reduces the machining time as well as distance between two machines also saves labour efforts for material handling.

V. CONCLUSION

The goal of this paper represents simple methods and modifications can be used for improving operations and work process in alloy wheel manufacturing. This study get the current methods used for manufacturing and how much time required for skilled operator to manufacture alloy wheel. By making some changes in the process, it can reduce the time taken for operations and save traveling distance by combining operations, it's also reduces material handling time in operations and labour efforts. The throughout in wheel manufacturing, total time is saves without compromising with quality and cost of the wheel.

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